Task 1. Nested Radicals

Available marks: 12

Nested radicals consist of *n*-th roots of an expression that includes another *n*-th root, nesting to infinity. The overall expression often converges quickly to an important quantity.

Write a program to evaluate and display two expressions based on nested radicals (in this case they're both square roots), accurate to 12 decimal places. That is, you should stop the evaluation when the absolute value of the difference between successive terms is 10^{-12} or less. The expressions are

$$A = \sqrt{1 + \sqrt{1 + \sqrt{1 + \sqrt{1 + \ldots}}}}$$

$$B = \sqrt{\tfrac{1}{2}} \times \sqrt{\tfrac{1}{2} + \tfrac{1}{2}\sqrt{\tfrac{1}{2}}} \times \sqrt{\tfrac{1}{2} + \tfrac{1}{2}\sqrt{\tfrac{1}{2} + \tfrac{1}{2}\sqrt{\tfrac{1}{2}}}} \times \ldots$$

The program should display the following four quantities:

$$A, \quad \frac{1}{1-A}, \quad B, \quad \frac{2}{B}$$

This task doesn't use a data file.

Assessment

When you believe you have completed this task (confirmed by patterns evident in your answers), show the judges the program and its output.

A is worth **5 marks**, B is worth **7 marks**.

Source: Wolfram Mathworld
mathworld.wolfram.com/NestedRadical.html